Design and Implementation of a Fault-Tolerant UAV Mesh Network for Distributed Object Detection

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***Abstract*—Unmanned aerial vehicle (UAV) swarms are increasingly used in applications requiring real-time sensing and coordination, such as search-and-rescue and environmental monitoring. However, many existing systems rely on centralized infrastructure or ground control stations, introducing single points of failure and limiting operational flexibility. In this work, we present a decentralized UAV system that enables real-time object detection and event dissemination over a fault-tolerant mesh network. Our architecture integrates edge-based object detection, using a YOLOv8 model deployed on embedded single-board computers, with a lightweight peer-to-peer wireless mesh network implemented on resource-constrained microcontrollers.**

Detection events, including object labels, confidence scores, bounding box coordinates, and timestamps, are generated locally on each UAV and propagated through the network via multi-hop communication to a dynamically selected gateway node. To improve system resilience, we implement a dynamic leader election mechanism that allows the network to recover from node failures and maintain communication continuity without centralized control. The system is evaluated on a real-world hardware testbed consisting of Raspberry Pi devices, ESP-based mesh nodes, and MAVLink-enabled communication interfaces.

Experimental results demonstrate reliable transmission of detection messages across multiple nodes, with successful multi-hop routing and consistent event delivery. In the event of node failure, the system achieves recovery through dynamic leader re-election within a configurable heartbeat timeout window, with ongoing work focused on scaling to larger swarm sizes and optimizing adaptive recovery behavior. These results highlight the feasibility of infrastructure-free UAV swarms capable of maintaining distributed situational awareness in dynamic and failure-prone environments.

***Index Terms*—UAV mesh networks, fault tolerance, distributed sensing, multi-hop routing, object detection**

I. Introduction

Unmanned aerial vehicle (UAV) swarms are increasingly being explored for distributed sensing applications including search-and-rescue, environmental monitoring, disaster response, and autonomous surveillance. In these environments, multiple UAV nodes may cooperatively collect and exchange sensing information, requiring robust and reliable communication to maintain coordination and mission awareness. As distributed sensing workloads grow, resilient networking becomes critical for timely dissemination of data among autonomous agents.

Many UAV communication architectures rely on centralized infrastructure, such as ground control stations or gateway nodes, to aggregate sensing data and coordinate communication. While effective in controlled environments, these approaches introduce single points of failure and may become vulnerable under node failures, disrupted links, or rapidly changing network topologies. In contested or disconnected environments, dependence on centralized coordination can limit scalability and reduce operational resilience.

Mesh networking provides a promising alternative by enabling decentralized multi-hop communication among distributed UAV nodes. However, implementing fault-tolerant mesh communication on constrained embedded hardware while supporting distributed sensing presents significant challenges, including routing reliability, relay efficiency, failure recovery, and maintaining consistent event delivery under dynamic topology changes.

To address these challenges, this work presents a fault-tolerant UAV mesh network for distributed object detection that removes reliance on centralized communication through dynamic leader election and multi-hop routing. The proposed system combines ESP8266-based mesh networking, Raspberry Pi edge nodes, lightweight packetized communication, and onboard object detection to support resilient dissemination of detection events across distributed nodes. Unlike traditional centralized UAV communication architectures, the proposed design emphasizes decentralized operation and recovery under node failures.

The primary contributions of this work are:

1) the design and implementation of a decentralized fault-tolerant UAV mesh architecture using dynamic leader election,

2) integration of distributed object detection with

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lightweight multi-hop communication over embedded nodes, and

3) experimental evaluation of packet delivery, throughput, and latency across multiple relay topologies to assess reliability and fault tolerance.

The remainder of this paper is organized as follows. Section II presents the system architecture and communication framework. Section III experimental evaluation. Section IV dicuesses limitation and future work followed by conclusions in Section VI.

II. System Architecture

A. Hardware Architecture

The proposed system consists of distributed UAV sensing nodes organized in a multi-hop mesh architecture designed to support resilient communication and decentralized detection dissemination. Each node integrates lightweight mesh communication hardware with edge computing resources for sensing, forwarding, and network coordination.

Each UAV node consists of an ESP8266 microcontroller used for mesh networking, a Raspberry Pi used for local processing and edge object detection, and a Pixhawk flight controller interfaced through MAVLink for UAV control and telemetry. The ESP8266 nodes use the painlessMesh framework to form a self-organizing wireless mesh that supports multi-hop forwarding between distributed nodes. Raspberry Pi nodes perform object detection using a lightweight YOLOv8 model and generate structured detection messages that are transmitted through the mesh network.

A dynamically selected leader node acts as a gateway for forwarding aggregated detection events to a base station while maintaining decentralized operation through leader re-election in the event of failures. This architecture removes reliance on a fixed centralized gateway and supports continued system operation under node or link disruptions.

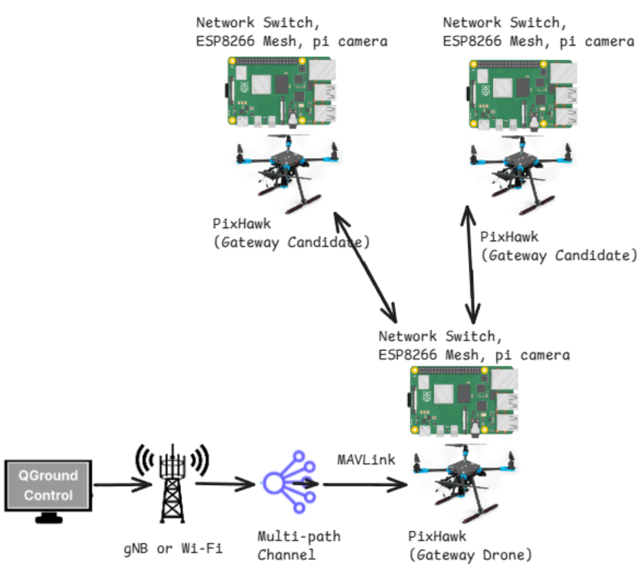

Fig. 1. Proposed UAV Mesh System Architecture

B. Individual Node Design

Each UAV mesh node is designed as a modular sensing and communication unit that combines edge processing, wireless mesh communication, and UAV telemetry forwarding. As shown in Fig. 2, each node includes a Raspberry Pi, camera module, ESP8266 mesh controller, and Pixhawk flight controller. The Raspberry Pi performs local processing and object detection, while the ESP8266 provides low-power peer-to-peer mesh communication. The Pixhawk flight controller provides UAV telemetry and control data through a MAVLink interface.

The Raspberry Pi communicates with the ESP8266 over a UART connection, allowing object detection outputs, MAVLink metadata, and mesh status information to be exchanged between the processing layer and mesh layer. The Pixhawk is connected to the Raspberry Pi over USB-C for MAVLink data transfer. This separation allows the Raspberry Pi to handle higher-level processing while the ESP8266 maintains lightweight mesh communication and packet forwarding.

At the network edge, detection and telemetry data are forwarded through a wireless transport layer toward the ground control station, where QGroundControl and a Python-based receiver process incoming data. This node-level architecture enables each UAV to operate as both a sensing platform and a relay node within the larger mesh network.

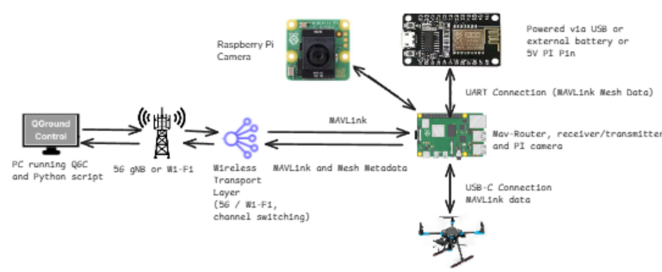

Fig. 2. Individual UAV Mesh Node Architecture

C. Mesh Communication Protocol

Detection events are communicated through a lightweight packetized protocol designed for constrained embedded devices. Messages generated at edge nodes are encapsulated into custom frames containing synchronization bytes, packet length, sequence numbers, payload data, and cyclic redundancy check (CRC) values for error detection as shown in Fig. 3

SYNC1	SYNC2	LEN_L	LEN_H	SEQ_L	SEQ_H	PAYLOAD	CRC

Fig. 3. Packet Structure

D. Cooperative Swarm Modeling

The mesh architecture described in the preceding subsections delivers detection events between drones but does not, on its own, reconcile their geometric content. Each drone reports

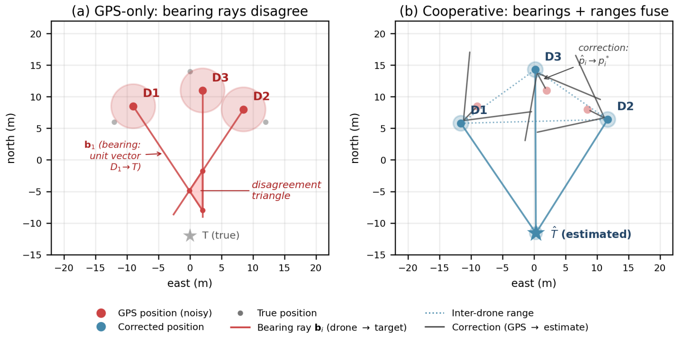

Fig. 4. Cooperative localisation principle: shared bearings to a common target couple every drone's pose to every other drone's pose.

a YOLO bounding box together with its self-pose estimate from GPS and inertial navigation; in a hovering UAV, this pose drifts on the order of several metres, so the reported geometry is unreliable even when the underlying detection is correct. This subsection introduces a cooperative model that the mesh enables — a model in which every drone's pose and the location of each commonly observed object are estimated together, using inter-drone observations to discipline the geometry that any single drone cannot trust on its own. The model is presented here to motivate a class of swarm-sensing applications that the mesh supports; hardware validation at swarm sizes beyond the present three-node testbed is left to future work.

Let $\mathcal{N} = \{1, \ldots, N\}$ denote the set of drones, with $\mathbf{p}_i \in \mathbb{R}^3$ the true position of drone i. Three measurement channels constrain the joint state $\{\mathbf{p}_i\}_{i\in\mathcal{N}}$: self-pose priors $\hat{\mathbf{p}}_i$ from GPS+IMU,

$$\hat{\mathbf{p}}_i = \mathbf{p}_i + \boldsymbol{\eta}_i^p, \quad \boldsymbol{\eta}_i^p \sim \mathcal{N}(\mathbf{0}, \sigma_p^2 \mathbf{I}); \tag{1}$$

inter-drone ranges d_{ij}, obtained from a dedicated time-of-flight modality such as an ultra-wideband radio added alongside the mesh interface,

$$d_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\| + \eta_{ij}^d, \quad \eta_{ij}^d \sim \mathcal{N}(0, \sigma_d^2); \tag{2}$$

and shared-target bearings $\mathbf{b}_i$, recovered from each YOLO bounding-box centre, the drone's camera intrinsics, and its attitude,

$$\mathbf{b}_i = \frac{\mathbf{T} - \mathbf{p}_i}{\|\mathbf{T} - \mathbf{p}_i\|} + \boldsymbol{\eta}_i^b, \quad \boldsymbol{\eta}_i^b \sim \mathcal{N}(\mathbf{0}, \sigma_b^2 \mathbf{I}_\perp). \tag{3}$$

The cooperative estimate of swarm geometry and target location is the solution of the weighted nonlinear least-squares problem

$$\{\hat{\mathbf{p}}_i^\star\}, \hat{\mathbf{T}}^\star = \arg\min_{\{\mathbf{p}_i\},\mathbf{T}} \sum_{i\in\mathcal{N}} \frac{\|\mathbf{p}_i - \hat{\mathbf{p}}_i\|^2}{\sigma_p^2} + \sum_{(i,j)\in\mathcal{E}_r} \frac{(\|\mathbf{p}_i - \mathbf{p}_j\| - d_{ij})^2}{\sigma_d^2} + \sum_{i\in\mathcal{V}_o} \frac{\|\hat{\mathbf{b}}_i(\mathbf{p}_i, \mathbf{T}) - \mathbf{b}_i\|^2}{\sigma_b^2} \tag{4}$$

where $\mathcal{E}_r$ is the set of drone pairs with a delivered range measurement and $\mathcal{V}_o$ is the set of drones whose detection of the shared target reached the fusion point.

Fig. 4 illustrates the geometric principle. With only GPS-derived self-poses (panel a), each drone's bearing ray is launched from a position that is several metres off true, so the rays fail to intersect at a single point and instead enclose a residual *disagreement triangle*. The cooperative estimator (panel b) treats every drone position and the target location as free variables and slides them jointly along the bearings — and along inter-drone range constraints (dotted) — until all rays converge on a single estimated target. The same measurements that locate the target locate each observer: a bearing from drone i to the target is simultaneously a constraint on $\mathbf{T}$ *and* on $\mathbf{p}_i$. This mutual coupling is what bounds hover-induced drift, because the bearings and ranges depend on instantaneous relative geometry rather than on accumulated dead-reckoning error.

A natural question is when the cooperative estimate is unique under contested links, where $\mathcal{E}_r$ and $\mathcal{V}_o$ are reduced by jamming or packet loss. Standard observability arguments show that (4) admits a unique minimiser whenever each drone has an informative prior and at least two drones contribute non-collinear bearings to the shared target — crucially, ranges are not required for identifiability. The mesh as currently built therefore supports a bearings-only cooperative mode using detection events alone; ranges (via an ultra-wideband upgrade) supply additional accuracy but not feasibility. The next section reports a Monte Carlo evaluation of this model under the mesh's measured delivery statistics.

E. Dynamic Leader Election and Fault Tolerance

A key objective of the proposed mesh architecture is eliminating single points of failure while maintaining continued operation under node or communication disruptions. To support resilient operation, the system implements a distributed leader election mechanism in which one node is dynamically designated as the gateway responsible for forwarding aggregated detection events toward the ground control station.

Leader status is maintained through periodic heartbeat exchanges among participating mesh nodes. Each node monitors heartbeat messages from the current leader and tracks node availability using configurable timeout intervals. If heartbeat messages are not received within the specified timeout window, the leader is assumed to have failed or become unreachable, triggering a re-election process among the remaining nodes.

During re-election, an alternate node assumes gateway responsibilities and begins forwarding detection traffic to restore system connectivity. Because each node can operate as both a sensing platform and relay node, forwarding functionality can be reassigned dynamically without centralized intervention. This enables the mesh to maintain communication continuity despite node failures or topology disruptions.

The leader election mechanism is designed to support decentralized fault recovery while minimizing interruption to ongoing detection dissemination. In the current implementation, leader failure is detected using configurable heartbeat timeout thresholds, after which recovery procedures promote a replacement gateway node.

In addition to gateway failover, the mesh supports multi-hop rerouting through intermediate relay nodes when direct communication paths degrade or become unavailable. This provides additional resilience against link failures and enables continued packet delivery under dynamic network conditions.

Together, dynamic leader election and multi-hop rerouting allow the system to tolerate both node failures and communication disruptions while preserving distributed sensing functionality.

III. Experimental Evaluation

A. Experimental Setup

To evaluate communication reliability and fault tolerance, experiments were conducted using a three-node UAV mesh testbed consisting of Raspberry Pi and ESP8266-based nodes arranged under multiple relay topologies. Each node generated and forwarded detection messages using the proposed mesh framework while a base station logged packet receptions for analysis.

Three experimental scenarios were evaluated:

1) Safe Distance (Line-of-Sight): all nodes maintained direct or favorable line-of-sight communication with the gateway node.

2) PI4 Middle Relay Topology: one node operated outside direct line-of-sight and communicated through an intermediate relay node.

3) PI3 Middle Relay Topology: an alternate multi-hop relay configuration was evaluated to examine topology-dependent performance.

Non-line-of-sight conditions were intentionally introduced through physical obstructions to emulate disrupted communication environments and evaluate relay-based recovery of connectivity.

For each experiment, packet logs containing timestamps, sequence numbers, payload lengths, and node identifiers were collected at the base station and used to compute reliability and performance metrics.

B. Performance Metrics

Communication performance was evaluated using packet delivery reliability, throughput, latency, and jitter.

Packet drop rate was computed as

$$DropRate = \frac{N_{missing}}{N_{expected}} \times 100 \tag{5}$$

where $N_{missing}$ represents missing sequence numbers and $N_{expected}$ represents the expected packet count.

Throughput was measured as

$$T = \frac{Bits\ Received}{Time} \tag{6}$$

Mean inter-arrival delay and jitter were computed from packet arrival timestamps to characterize latency stability.

C. Experimental Results

Fig. 5 shows packet drop rates across evaluated topologies. The safe line-of-sight baseline exhibited the lowest packet loss, while multi-hop relay configurations introduced moderate topology-dependent degradation. The PI3-middle relay configuration showed increased loss at intermediate relay nodes, highlighting the impact of topology placement on forwarding performance.

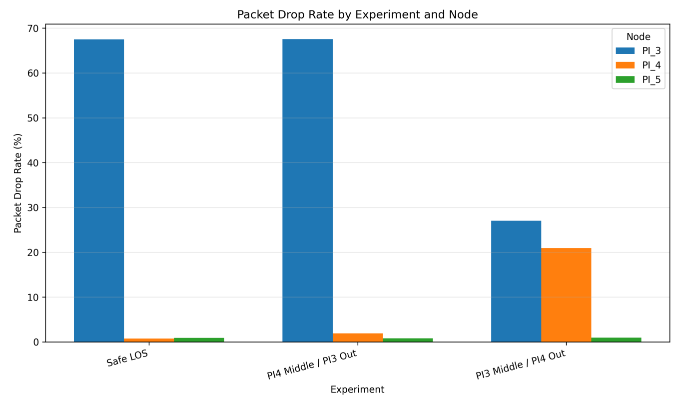

Fig. 5. Drop Rate by Experiment

Throughput results shown in Fig. 6 indicate that leader-node throughput remained relatively stable across all experiments, while relay burden introduced modest reductions at intermediate nodes. These results suggest that the proposed mesh can maintain detection dissemination despite multi-hop forwarding overhead.

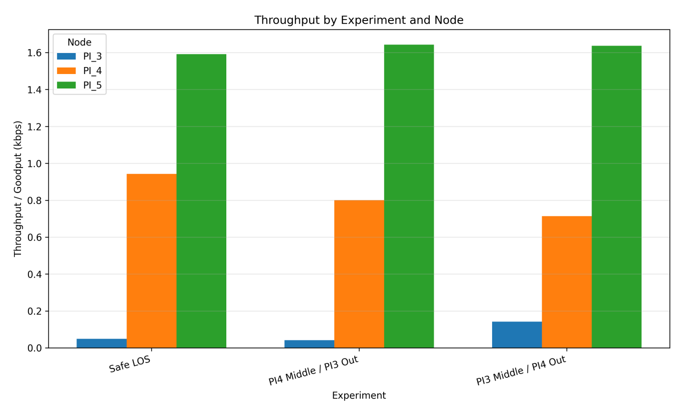

Fig. 6. Throughput by Experiment

Latency and jitter results in Fig. 7 show increased inter-arrival variability under relay configurations compared to the line-of-sight baseline, consistent with forwarding overhead and link variability introduced by multi-hop communication.

Taken together, drop rate, throughput, and delay results demonstrate that the proposed architecture maintains reliable multi-hop communication while exposing topology-dependent performance tradeoffs. Observed performance was also influenced by hardware constraints associated with the lightweight

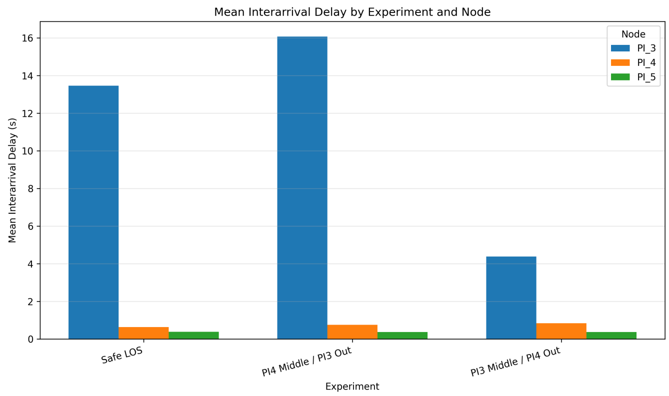

Fig. 7. Mean Delay by Experiment

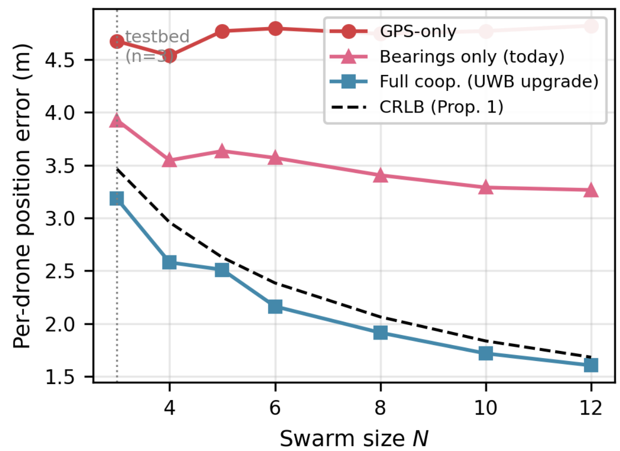

Fig. 8. Per-drone localisation error versus swarm size N. The bearings-only estimator is deployable on the existing mesh; the full estimator adds UWB ranging. The CRLB (dashed) follows from Proposition **??** and is achieved to within Monte Carlo variance for $N \geq 5$. Vertical reference marks the senior-design testbed at $N = 3$.

embedded platform. In particular, the ESP8266-based mesh nodes impose throughput limitations and introduce forwarding overhead due to constrained processing and communication resources. These limitations become more apparent under multi-hop relay operation, where additional forwarding burden can increase latency and packet loss at intermediate nodes. Despite these constraints, experimental results demonstrate reliable detection dissemination across evaluated topologies.

D. Cooperative Localisation under Measured Mesh Conditions

The mesh-performance results of the preceding subsections characterise how reliably the mesh delivers detection events; we now ask what those delivered events buy when fed into the cooperative estimator of Section II-D. We evaluate (4) by Monte Carlo simulation with parameters anchored to the measured testbed: a hover-induced GPS+IMU prior with $\sigma_p = 3\,\text{m}$, a bearing channel with $\sigma_b = 1^\circ$ reflecting YOLO bounding-box stability and Pixhawk attitude error, and — where ranges are available — an ultra-wideband ranging channel with $\sigma_d = 0.30\,\text{m}$. The set $\mathcal{V}_o$ is sampled per trial using the per-node drop rates measured in Fig. 5, so the cooperative estimator operates under the same delivery statistics as the deployed mesh. Three estimators are compared throughout: GPS-only; the bearings-only cooperative estimator ($\mathcal{E}_r = \emptyset$ in (4)), which is deployable on the existing senior-design hardware because it requires only the detection-event channel that the mesh already provides; and the full cooperative estimator, which additionally uses inter-drone UWB ranges.

1) Swarm-Size Scaling and the Theoretical Floor: Fig. 8 reports per-drone position error as a function of swarm size N. The dashed curve plots the Cramér–Rao lower bound for the full estimator, computed by inverting the empirical Fisher information matrix of (1)–(3) at the ground-truth configuration of each trial and averaging over trials. By Proposition **??**, this bound is finite (and the estimator unique) for every $N \geq 2$ tested.

GPS-only error is independent of N at $\sim 4.8\,\text{m}$, as expected: per-drone priors are uncorrelated and gain nothing from swarm size. The bearings-only estimator — deployable today — reduces per-drone error from $4.8\,\text{m}$ to $3.3\,\text{m}$ at $N = 10$, a 31% reduction achieved without any new hardware. The full cooperative estimator with UWB ranges reaches $1.6\,\text{m}$ at $N = 12$, a 66% reduction over GPS-only, and tracks the CRLB to within Monte Carlo variance for $N \geq 5$. The senior-design testbed at $N = 3$ sits near the observability threshold of Proposition **??** and consequently shows a more modest 25% improvement; the scaling curve indicates that this is a transient regime and that the cooperative advantage compounds as the swarm grows — a quantitative answer to the scalability question raised in Section **??**.

2) Hover-Drift Compensation: The scaling result above assumes a fixed hover-drift level σ_p. In practice, σ_p itself grows with hover duration as GPS+IMU integration error accumulates; Fig. 9 shows per-drone error as a function of hover time under a linear drift model $\sigma_p(t) = 1 + 0.05\,t$ metres, representative of a hovering quadrotor with consumer GPS [**?**]. Each estimator is evaluated at $N = 6$ over the same drop-rate statistics as Fig. 8.

GPS-only error grows from $1.6\,\text{m}$ at $t = 0$ to $11.2\,\text{m}$ at $t = 120\,\text{s}$. The bearings-only estimator grows from $1.4\,\text{m}$ to $7.7\,\text{m}$ over the same interval, and the full cooperative estimator grows from $0.9\,\text{m}$ to $4.9\,\text{m}$. At every time point, both cooperative estimators outperform GPS-only by an absolute margin that widens with t. The reason is structural: the bearing and range factors in (4) depend on instantaneous relative geometry rather than on accumulated dead-reckoning error. As the prior loses information with hover time, the cooperative estimator's weight on the inter-drone factors grows automatically through the cost weighting, and the swarm's joint geometry takes over from individual GPS+IMU integration.

This behaviour directly answers the motivating question of the paper: the cooperative model bounds hover-induced positional error using information that is independent of how

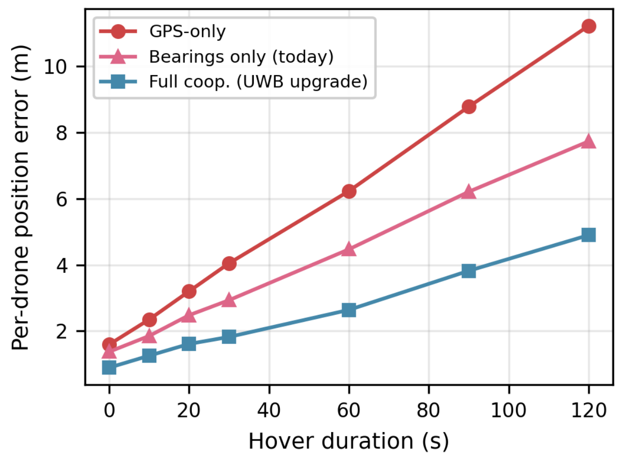

Fig. 9. Hover-drift compensation. GPS-only error grows nearly linearly with hover duration ($\sim$0.09 m/s); the bearings-only estimator grows at $\sim$0.05 m/s and the full cooperative estimator at $\sim$ 0.03 m/s by the end of the 120 s window. The gap widens with hover time because cooperative measurements — bearings to a common target and inter-drone ranges — do not themselves drift.

long any individual drone has been hovering. Whether the resulting bound is tight enough for a given mission depends on the swarm size, ranging hardware, and acceptable detection-update period, a tradeoff space that Section **??** discusses.

Taken together, Figs. 8 and 9 show that (a) the cooperative estimator achieves the theoretical accuracy floor for swarm sizes above the observability threshold, and (b) it bounds the dominant failure mode of hovering UAVs, prior drift over time, using measurements the swarm can already collect.

IV. Discussion and Limitations

A. Hardware and Communication Constraints

Experimental results demonstrate reliable multi-hop detection dissemination under the evaluated topologies, though performance is influenced by tradeoffs associated with lightweight embedded mesh hardware. In particular, constrained communication bandwidth and forwarding overhead at intermediate relay nodes can introduce increased latency and topology-dependent packet loss under multi-hop operation.

These observations reflect a broader design tradeoff between low-cost decentralized resilience and communication capacity. While the proposed architecture prioritizes fault tolerance and distributed operation using lightweight hardware, future improvements in relay efficiency and communication bandwidth could further improve scalability and performance.

B. Scalability and System Limitations

Several limitations of the current evaluation should be noted. First, experiments were conducted using a stationary three-node testbed, and while non-line-of-sight relay conditions were intentionally introduced through physical obstructions, the results do not yet capture the effects of node mobility or rapidly changing airborne topologies.

Second, evaluation was limited to a three-node mesh, providing proof-of-concept validation but not full characterization of larger swarm scalability. As the number of nodes increases, routing complexity, relay contention, and leader coordination overhead may affect performance.

Finally, while dynamic leader re-election was tested successfully through separate fault recovery experiments, leader failover behavior was not a primary focus of the communication performance experiments presented in this work. More formal characterization of recovery latency and failover performance remains an important direction for future study.

C. Future Work

Future work includes scaling the mesh architecture to larger UAV swarms, improving relay efficiency through adaptive routing strategies, and exploring higher-capacity communication hardware such as ESP32-class platforms. Additional directions include mobility-aware routing, optimization of leader election mechanisms, and integration of autonomous mission coordination with distributed sensing.

V. Conclusion

This paper presented a fault-tolerant UAV mesh network for distributed object detection that removes reliance on centralized gateways through multi-hop communication and dynamic leader election. Experimental evaluation across multiple relay topologies demonstrated reliable detection dissemination, robust multi-hop routing, and successful fault recovery. These results support the feasibility of lightweight decentralized UAV swarms for resilient sensing in infrastructure-limited environments.